Experimental Design – EMS Readiness Simulation Study

1. Research Questions and Hypotheses

Primary Research Questions

ID	Research Question
RQ1	How do the three allocation policies (P0: Uniform, P1: Demand-Proportional, P2: Demand-Weighted Optimized) compare in terms of mean response time, P90 (90th percentile) response time, 6-minute coverage (NYC law), and 8-minute coverage (NFPA standard)?
RQ2	How sensitive is each policy's performance to fleet size (K)?
RQ3	How robust is each policy to changes in demand intensity?
RQ4	How do service time variations affect policy performance?

Hypotheses

ID	Hypothesis	Rationale
H1	P2 achieves lower mean response time than P1, which outperforms P0	Optimization-based allocation better matches supply to demand
H2	P2's advantage over P0 diminishes as K increases	With excess capacity, allocation quality matters less
H3	All policies degrade under high demand, but P0 degrades fastest	Uniform allocation wastes units in low-demand areas
H4	P2 is more robust to service time variation than P0	Optimized placement compensates for longer on-scene times

2. Experimental Factors and Levels

Factor Summary

Factor	Symbol	Levels	Values
Allocation Policy	π	3	P0 (Uniform), P1 (Demand-Proportional), P2 (Demand-Weighted)
Fleet Size	K	6	15, 20, 25, 30, 35, 40
Demand Multiplier	δ	6	0.5, 0.75, 1.0, 1.25, 1.5, 2.0
Service Time Mean	μ_s	3	20 min (−20%), 25 min (baseline), 30 min (+20%)

Capacity Constraint

The **firehouse capacity** parameter (C) defines the maximum number of EMS units that can be staged at any single firehouse. The operational default is **C = 2 units per firehouse**, established by the Capacity Sensitivity experiment (see `docs/capacity_sensitivity_analysis.md`).

At fleet sizes $K \leq 30$, the capacity constraint does not bind—results are identical whether C = 2 or C = 5. This was confirmed by the full-spectrum capacity sensitivity analysis testing cap = 1 through 5.

> **Historical note:** The experiments in Section 4 below (Exp1–Exp4) were originally designed with C = 5. All production results reported in the technical report (§5) use C = 2, which produces numerically identical outcomes at the fleet sizes tested. See the technical report §4.5.1 for the consolidated experimental design table.

Factor Details

Allocation Policies

- **P0 (Uniform)**: Equal distribution of K units across active firehouses (round-robin remainder)
- **P1 (Demand-Proportional)**: Units proportional to nearest-firehouse demand credit
- **P2 (Demand-Weighted Optimized)**: MIP minimizing expected demand-weighted response time

Fleet Size (K)

Pre-computed allocations at $K \in \{20, 30, 40, 48\}$. For intermediate values (15, 25, 35), allocations are generated dynamically using the optimization and baseline policy functions.

Demand Scaling

Applied by multiplying the NHPP base arrival rate (3.48 crashes/hour) by δ. All temporal patterns (hourly, day-of-week) and spatial patterns (precinct probabilities) remain unchanged.

Service Time Scenarios

The baseline lognormal service time (μ = 25 min, σ = 10 min) is varied by adjusting the mean:
- Low: μ = 20 min (−20%)

- Baseline: $\mu = 25$ min
- High: $\mu = 30$ min (+20%)

Standard deviation is held proportional (σ/μ ratio constant).

3. Response Variables

Metric	Symbol	Description
Mean Response Time	$E[RT]$	Average time from incident arrival to unit arrival on scene
P90 (90th %ile) RT	RT_{90}	90th percentile response time — exceeded by only 10% of incidents
95th Percentile RT	RT_{95}	Response time exceeded by only 5% of incidents
8-min Coverage	C_8	Fraction of incidents with response time ≤ 8 min
10-min Coverage	C_{10}	Fraction of incidents with response time ≤ 10 min
Mean Utilization	$\bar{\rho}$	Average unit utilization across fleet
Max Utilization	ρ_{max}	Maximum single-unit utilization
Mean Queue Length	$\bar{L}$	Time-weighted average queue length
Max Queue Length	L_{max}	Peak queue length observed
Queue Fraction	Q_{frac}	Fraction of incidents that waited in queue
Total Incidents	N	Total incidents generated
Incidents Queued	N_q	Number of incidents that entered queue

4. Experiment Sets

Experiment 1: Baseline Policy Comparison

- **Objective**: Compare P0, P1, P2 at baseline conditions
- **Factors**: $\pi \in \{P0, P1, P2\}$, $K = 20$, $\delta = 1.0$, $\mu_s = 25$, **C = 5**
- **Runs**: 3 policies × 30 replications = **90 runs**
- **Output**: `results/simulation/production/exp1_policy_comparison.csv`

Experiment 2: Fleet Size Sensitivity

- **Objective**: Evaluate performance across fleet sizes
- **Factors**: $\pi \in \{P0, P1, P2\}$, $K \in \{15, 20, 25, 30, 35, 40\}$, $\delta = 1.0$, $\mu_s = 25$, **C = 5**
- **Runs**: 3 × 6 × 30 = **540 runs**
- **Output**: `results/simulation/production/exp2_fleet_sensitivity.csv`

Experiment 3: Demand Scaling Sensitivity

- **Objective**: Assess robustness to demand changes
- **Factors**: $\pi \in \{P0, P1, P2\}$, $K = 20$, $\delta \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$, $\mu_s = 25$, **C = 5**
- **Runs**: 3 × 6 × 30 = **540 runs**
- **Output**: `results/simulation/production/exp3_demand_sensitivity.csv`

Experiment 4: Service Time Robustness

- **Objective**: Test sensitivity to on-scene service duration
- **Factors**: $\pi \in \{P0, P1, P2\}$, $K = 20$, $\delta = 1.0$, $\mu_s \in \{20, 25, 30\}$, **C = 5**
- **Runs**: 3 × 3 × 30 = **270 runs**
- **Output**: `results/simulation/production/exp4_service_robustness.csv`

Total Experimental Runs

Experiment	Runs
Exp 1: Policy Comparison	90
Exp 2: Fleet Sensitivity	540
Exp 3: Demand Sensitivity	540
Exp 4: Service Robustness	270
Total	**1,440**

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5. Replication Strategy

Common Random Numbers (CRN)

To reduce variance across policy comparisons, all policies within the same experiment/scenario share the same random number seeds per replication:
- Replication `i` uses seed `seed_base + i` where `seed_base = 42`

- Seeds: 42, 43, 44, …, 71 (for 30 replications)
- This ensures the same incident arrival pattern for all policies at the same replication number

Replication Count Justification

- **30 replications** per scenario provides approximately ±5% half-width for 95% confidence intervals on mean response time
- Validated in Phase 4 pilot runs showing stable CI widths

Simulation Horizon

- **168 hours** (1 week) per replication
- No warm-up period (terminating simulation)
- Captures full weekly demand cycle (hourly + day-of-week patterns)

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6. Output Data Structure

Each experiment CSV contains one row per replication with columns:

```
experiment_id - Experiment identifier (exp1, exp2, exp3, exp4)
scenario_id - Unique scenario label
replication - Replication number (0-29)
policy - Policy name (P0, P1, P2)
K - Fleet size
demand_multiplier - Demand scaling factor
service_time_mean - Service time mean (minutes)
mean_response_time - Mean response time (minutes)
p90_response_time - P90 (90th percentile) response time (minutes)
coverage_6min - fraction of calls with response time ≤ 6 minutes (NYC law)
p95_response_time - 95th percentile response time (minutes)
coverage_8min - Fraction with RT ≤ 8 min
coverage_10min - Fraction with RT ≤ 10 min
mean_utilization - Mean fleet utilization
max_utilization - Maximum unit utilization
mean_queue_length - Time-weighted average queue length
max_queue_length - Peak queue length
queue_fraction - Fraction of incidents queued
total_incidents - Total incidents in replication
incidents_queued - Number queued
random_seed - Random seed used
```

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7. Analysis Plan

Statistical Methods

1. **Paired t-tests** (using CRN) for pairwise policy comparison
2. **ANOVA** for multi-factor analysis across fleet sizes
3. **95% confidence intervals** for all point estimates
4. **Effect size** (Cohen’s d) for practical significance

Visualizations

1. Box plots: response time distributions by policy

2. Line plots: performance metrics vs. fleet size (sensitivity curves)
3. Heatmaps: policy × demand multiplier performance
4. Confidence interval plots for key comparisons

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